

Topic 06: Laplace Transform of Periodic Functions

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

The AC power supply in your home delivers a sinusoidal voltage repeating 50 times per second. A digital clock circuit uses a square wave. A sawtooth wave drives the sweep in old CRT television sets. All of these are *periodic functions* — they repeat with a fixed period T . Computing their Laplace transforms from scratch using the definition integral would require handling an infinite sum of integrals. There is a much smarter formula that condenses everything into a single integral over just ONE period. Let's derive it.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

Engineering systems are driven by periodic inputs: square waves in digital circuits, sawtooth waves in power electronics, triangular waves in PWM controllers. Without the periodic function formula, computing $\mathcal{L}\{f(t)\}$ for even a simple square wave requires summing an infinite series of separate integrals — completely impractical under exam conditions.

Mistake	Consequence
Using the definition integral directly for periodic functions	Infinitely tedious computation — cannot be completed in exam time
Forgetting the factor $1/(1 - e^{-sT})$	Computing only the first-period integral and losing the periodic scaling
Misidentifying the period T	Using the wrong limits in the formula

Key Takeaway

The periodic function theorem reduces an infinite sum of integrals to a single integral over one period, multiplied by a geometric-series factor. This is the single most powerful shortcut for signals in electrical engineering.

3 You Already Know This (Intuition First)

Imagine you receive \$100 every month (period $T = 1$ month) and the discount rate per period is s . The present value of all future payments is:

$$100 + 100 e^{-s} + 100 e^{-2s} + \dots = \frac{100}{1 - e^{-s}}.$$

This is a geometric series with ratio $e^{-s} < 1$.

The Laplace transform of a periodic function works *identically*:

- Integrate $e^{-st} f(t)$ over just the first period $[0, T]$ to get a “one-period payment”.
- Multiply by the geometric series factor $\frac{1}{1 - e^{-sT}}$ to account for all future repetitions.

The formula is literally a financial present-value calculation applied to waveforms.

4 Definition of Periodic Function

Definition — Periodic Function

A function $f(t)$ is said to be **periodic** with period $T > 0$ if

$$f(t + T) = f(t) \quad \text{for all } t \geq 0.$$

The smallest positive T satisfying this equation is called the **fundamental period**.

Function	Fundamental Period
$\sin(\omega t)$	$2\pi/\omega$
$\cos(\omega t)$	$2\pi/\omega$
Square wave (amplitude A)	T (given explicitly)
Triangular wave	T (given explicitly)
Sawtooth wave	T (given explicitly)
$ \sin t $ (full-wave rectified)	π

5 The Periodic Function Theorem

Theorem and Complete Proof

Theorem. If $f(t)$ is periodic with period $T > 0$ and piecewise continuous on $[0, T]$, then for $s > 0$:

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

Proof.

Step 1 — Split the definition integral over successive periods.

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt = \sum_{n=0}^\infty \int_{nT}^{(n+1)T} e^{-st} f(t) dt.$$

Step 2 — Substitute $t = u + nT$ in the n -th integral.

Let $t = u + nT$, so $dt = du$. When $t = nT$, $u = 0$; when $t = (n+1)T$, $u = T$. Therefore:

$$\int_{nT}^{(n+1)T} e^{-st} f(t) dt = \int_0^T e^{-s(u+nT)} f(u + nT) du.$$

Step 3 — Apply periodicity: $f(u + nT) = f(u)$.

$$= \int_0^T e^{-s(u+nT)} f(u) du = e^{-snT} \int_0^T e^{-su} f(u) du.$$

Step 4 — Sum over all $n \geq 0$.

Denote $I_1 = \int_0^T e^{-su} f(u) du$ (the one-period integral). Then:

$$\mathcal{L}\{f(t)\} = \sum_{n=0}^\infty e^{-snT} \cdot I_1 = I_1 \sum_{n=0}^\infty (e^{-sT})^n.$$

Step 5 — Evaluate the geometric series.

Since $s > 0$ we have $|e^{-sT}| < 1$, so the series converges:

$$\sum_{n=0}^{\infty} (e^{-sT})^n = \frac{1}{1 - e^{-sT}}.$$

Step 6 — Combine.

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

What Did We Learn?

The periodic function theorem converts an infinite sum of integrals into one integral over $[0, T]$ multiplied by the geometric-series factor $1/(1 - e^{-sT})$. The convergence of the geometric series requires $s > 0$, which is always satisfied in practice.

6 Application 1 — Square Wave

Square Wave: Derivation

Definition. The square wave with amplitude A and half-period a is defined by:

$$f(t) = \begin{cases} A & 0 \leq t < a \\ -A & a \leq t < 2a \end{cases}, \quad f(t + 2a) = f(t).$$

The fundamental period is $T = 2a$.

Step 1 — Write the one-period integral.

$$\int_0^{2a} e^{-st} f(t) dt = A \int_0^a e^{-st} dt - A \int_a^{2a} e^{-st} dt.$$

Step 2 — Evaluate each integral.

$$A \int_0^a e^{-st} dt = A \left[-\frac{e^{-st}}{s} \right]_0^a = \frac{A}{s} (1 - e^{-as}).$$

$$A \int_a^{2a} e^{-st} dt = A \left[-\frac{e^{-st}}{s} \right]_a^{2a} = \frac{A}{s} (e^{-as} - e^{-2as}).$$

Step 3 — Combine.

$$\int_0^{2a} e^{-st} f(t) dt = \frac{A}{s} (1 - e^{-as}) - \frac{A}{s} (e^{-as} - e^{-2as}) = \frac{A}{s} (1 - 2e^{-as} + e^{-2as}) = \frac{A}{s} (1 - e^{-as})^2.$$

Step 4 — Apply the theorem with $T = 2a$.

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-2as}} \cdot \frac{A}{s} (1 - e^{-as})^2.$$

Step 5 — Factor the denominator and simplify.

$$1 - e^{-2as} = (1 - e^{-as})(1 + e^{-as}).$$

$$\mathcal{L}\{f(t)\} = \frac{A}{s} \cdot \frac{(1 - e^{-as})^2}{(1 - e^{-as})(1 + e^{-as})} = \frac{A}{s} \cdot \frac{1 - e^{-as}}{1 + e^{-as}}.$$

Step 6 — Write in tanh form.

Multiply numerator and denominator of $\frac{1 - e^{-as}}{1 + e^{-as}}$ by $e^{as/2}$:

$$\frac{1 - e^{-as}}{1 + e^{-as}} = \frac{e^{as/2} - e^{-as/2}}{e^{as/2} + e^{-as/2}} = \tanh\left(\frac{as}{2}\right).$$

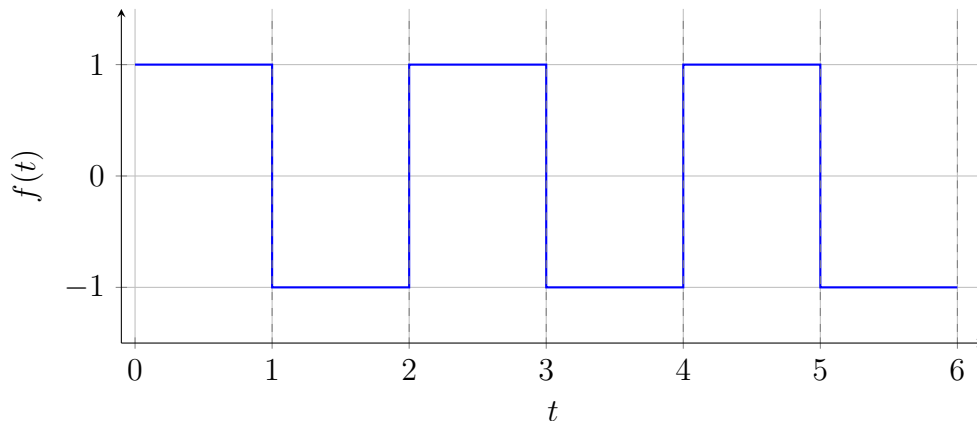
$$\boxed{\mathcal{L}\{\text{square wave}\} = \frac{A}{s} \tanh\left(\frac{as}{2}\right)}.$$

What Did We Learn?

The Laplace transform of a symmetric square wave with amplitude A and half-period a is $\frac{A}{s} \tanh\left(\frac{as}{2}\right)$. The tanh function appears naturally from the ratio $(1 - e^{-as})/(1 + e^{-as})$.

Square Wave — pgfplots Graph

Square Wave ($a = 1$, $A = 1$) over 3 periods



7 Application 2 — Triangular Wave

Triangular Wave: Derivation

Definition. The triangular wave with half-period a is:

$$f(t) = \begin{cases} t & 0 \leq t < a \\ 2a - t & a \leq t < 2a \end{cases}, \quad f(t + 2a) = f(t), \quad T = 2a.$$

Step 1 — One-period integral.

$$\int_0^{2a} e^{-st} f(t) dt = \int_0^a t e^{-st} dt + \int_a^{2a} (2a - t) e^{-st} dt.$$

Step 2 — Evaluate $I_A = \int_0^a t e^{-st} dt$ by parts.

Let $u = t$, $dv = e^{-st} dt$; then $du = dt$, $v = -e^{-st}/s$:

$$I_A = \left[-\frac{t e^{-st}}{s} \right]_0^a + \frac{1}{s} \int_0^a e^{-st} dt = -\frac{a e^{-as}}{s} + \frac{1}{s} \left[-\frac{e^{-st}}{s} \right]_0^a = -\frac{a e^{-as}}{s} + \frac{1 - e^{-as}}{s^2}.$$

Step 3 — Evaluate $I_B = \int_a^{2a} (2a - t) e^{-st} dt$.

Substitute $t = a + v$, so $2a - t = a - v$ and $dt = dv$; limits 0 to a :

$$I_B = \int_0^a (a - v) e^{-s(a+v)} dv = e^{-as} \int_0^a (a - v) e^{-sv} dv.$$

Apply parts with $u = a - v$, $dv' = e^{-sv} dv$:

$$\int_0^a (a - v) e^{-sv} dv = \left[-\frac{(a - v) e^{-sv}}{s} \right]_0^a - \int_0^a \frac{e^{-sv}}{s} dv = \frac{a}{s} - \frac{1}{s} \left[-\frac{e^{-sv}}{s} \right]_0^a = \frac{a}{s} - \frac{1 - e^{-as}}{s^2}.$$

Thus:

$$I_B = e^{-as} \left(\frac{a}{s} - \frac{1 - e^{-as}}{s^2} \right) = \frac{a e^{-as}}{s} - \frac{e^{-as}(1 - e^{-as})}{s^2}.$$

Step 4 — Combine $I_A + I_B$.

$$I_A + I_B = -\frac{a e^{-as}}{s} + \frac{1 - e^{-as}}{s^2} + \frac{a e^{-as}}{s} - \frac{e^{-as} - e^{-2as}}{s^2} = \frac{1 - 2e^{-as} + e^{-2as}}{s^2} = \frac{(1 - e^{-as})^2}{s^2}.$$

Step 5 — Apply the theorem.

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-2as}} \cdot \frac{(1 - e^{-as})^2}{s^2} = \frac{(1 - e^{-as})^2}{s^2(1 - e^{-as})(1 + e^{-as})} = \frac{1 - e^{-as}}{s^2(1 + e^{-as})}.$$

Multiplying numerator and denominator by $e^{as/2}$:

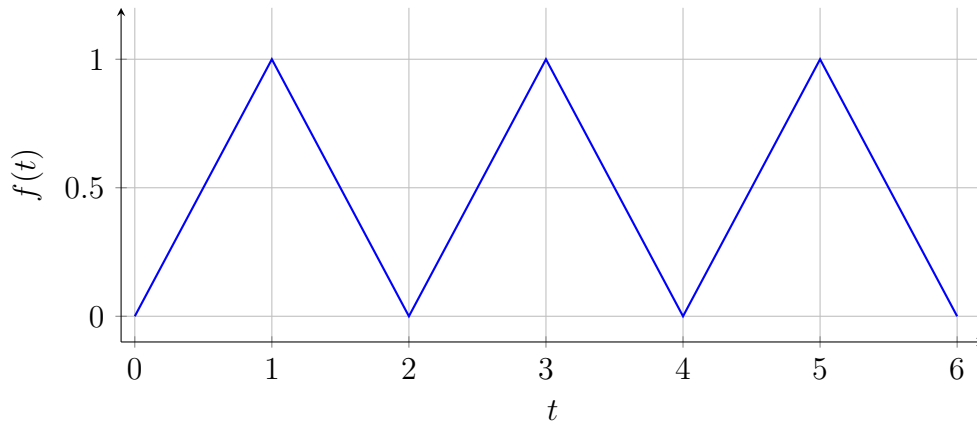
$$\boxed{\mathcal{L}\{\text{triangular wave}\} = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right).}$$

What Did We Learn?

The Laplace transform of the triangular wave with half-period a is $\frac{1}{s^2} \tanh\left(\frac{as}{2}\right)$. Notice the extra $1/s$ factor compared to the square wave — consistent with the triangular wave being an “integrated” square wave.

Triangular Wave — pgfplots Graph

Triangular Wave ($a = 1$) over 3 periods



8 Application 3 — Half-Wave Rectifier

Half-Wave Rectified Sine: Derivation

Definition.

$$f(t) = \begin{cases} \sin t & 0 \leq t < \pi \\ 0 & \pi \leq t < 2\pi \end{cases}, \quad T = 2\pi.$$

Step 1 — One-period integral.

$$\int_0^{2\pi} e^{-st} f(t) dt = \int_0^{\pi} e^{-st} \sin t dt.$$

Step 2 — Evaluate $\int_0^{\pi} e^{-st} \sin t dt$ by integration by parts twice.

Let $I = \int_0^{\pi} e^{-st} \sin t dt$. Apply parts: $u = \sin t$, $dv = e^{-st} dt$:

$$I = \left[-\frac{e^{-st} \sin t}{s} \right]_0^{\pi} + \frac{1}{s} \int_0^{\pi} e^{-st} \cos t dt.$$

The boundary term is zero (since $\sin 0 = \sin \pi = 0$), so:

$$I = \frac{1}{s} \int_0^{\pi} e^{-st} \cos t dt.$$

Apply parts again: $u = \cos t$, $dv = e^{-st} dt$:

$$\int_0^{\pi} e^{-st} \cos t dt = \left[-\frac{e^{-st} \cos t}{s} \right]_0^{\pi} - \frac{1}{s} \int_0^{\pi} e^{-st} \sin t dt = \frac{-e^{-\pi s}(-1) + e^0(1)}{s} - \frac{I}{s} = \frac{1 + e^{-\pi s}}{s} - \frac{I}{s}.$$

Substituting back:

$$I = \frac{1}{s} \left(\frac{1 + e^{-\pi s}}{s} - \frac{I}{s} \right) = \frac{1 + e^{-\pi s}}{s^2} - \frac{I}{s^2}.$$

$$I \left(1 + \frac{1}{s^2} \right) = \frac{1 + e^{-\pi s}}{s^2} \implies I \cdot \frac{s^2 + 1}{s^2} = \frac{1 + e^{-\pi s}}{s^2} \implies I = \frac{1 + e^{-\pi s}}{s^2 + 1}.$$

Step 3 — Apply the theorem.

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-2\pi s}} \cdot \frac{1 + e^{-\pi s}}{s^2 + 1} = \frac{1 + e^{-\pi s}}{(1 - e^{-2\pi s})(s^2 + 1)}.$$

Factor $1 - e^{-2\pi s} = (1 - e^{-\pi s})(1 + e^{-\pi s})$:

$$\mathcal{L}\{\text{half-wave rectified } \sin t\} = \frac{1}{(1 - e^{-\pi s})(s^2 + 1)}.$$

What Did We Learn?

For the half-wave rectified sine only the interval $[0, \pi]$ contributes to the one-period integral; the zero half collapses the upper limit to π . The result $1/[(1 - e^{-\pi s})(s^2 + 1)]$ neatly factors into a periodic scaling term and the sine transform $1/(s^2 + 1)$.

9 Worked Examples

Example 1 — Sawtooth Wave

Problem. A sawtooth wave is defined by $f(t) = t/T$ for $0 \leq t < T$, with period T . Find $\mathcal{L}\{f(t)\}$.

Step 1 — One-period integral.

$$\int_0^T e^{-st} \frac{t}{T} dt = \frac{1}{T} \int_0^T t e^{-st} dt.$$

Step 2 — Evaluate $\int_0^T t e^{-st} dt$ by parts.

$u = t$, $dv = e^{-st} dt \Rightarrow du = dt$, $v = -e^{-st}/s$:

$$\int_0^T t e^{-st} dt = \left[-\frac{t e^{-st}}{s} \right]_0^T + \frac{1}{s} \int_0^T e^{-st} dt = -\frac{T e^{-sT}}{s} + \frac{1}{s} \left[\frac{-e^{-st}}{s} \right]_0^T = -\frac{T e^{-sT}}{s} + \frac{1 - e^{-sT}}{s^2}.$$

Step 3 — Combine.

$$\frac{1}{T} \int_0^T t e^{-st} dt = \frac{1}{T} \left(-\frac{T e^{-sT}}{s} + \frac{1 - e^{-sT}}{s^2} \right) = -\frac{e^{-sT}}{s} + \frac{1 - e^{-sT}}{Ts^2}.$$

Step 4 — Apply the theorem.

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \left(-\frac{e^{-sT}}{s} + \frac{1 - e^{-sT}}{Ts^2} \right) = \frac{-e^{-sT}}{s(1 - e^{-sT})} + \frac{1}{Ts^2}.$$

Rewrite the first term by noting $\frac{-e^{-sT}}{1 - e^{-sT}} = \frac{e^{-sT}}{e^{-sT} - 1}$:

$$\mathcal{L}\{f(t)\} = \frac{1}{Ts^2} - \frac{e^{-sT}}{s(1 - e^{-sT})}.$$

Equivalently, multiplying numerator and denominator of the second term by e^{sT} :

$$\mathcal{L}\{f(t)\} = \frac{1}{Ts^2} - \frac{1}{s(e^{sT} - 1)}.$$

What Did We Learn?

For the unit-slope sawtooth wave with period T , the transform is $\frac{1}{Ts^2} - \frac{1}{s(e^{sT} - 1)}$. The $1/s^2$ factor is expected because sawtooth is a ramp function repeated periodically.

Example 2 — Full-Wave Rectified Sine

Problem. The full-wave rectified sine is $f(t) = |\sin t|$, which has period $T = \pi$. Find $\mathcal{L}\{f(t)\}$.

Step 1 — Note the period. On $[0, \pi]$: $|\sin t| = \sin t$ (since $\sin t \geq 0$ there).

$$\int_0^\pi e^{-st} |\sin t| dt = \int_0^\pi e^{-st} \sin t dt.$$

Step 2 — Evaluate (same integral as in the half-wave derivation).

$$\int_0^\pi e^{-st} \sin t dt = \frac{1 + e^{-\pi s}}{s^2 + 1}.$$

(Derived in Application 3 above.)

Step 3 — Apply the theorem with $T = \pi$.

$$\mathcal{L}\{|\sin t|\} = \frac{1}{1 - e^{-\pi s}} \cdot \frac{1 + e^{-\pi s}}{s^2 + 1} = \frac{1 + e^{-\pi s}}{(1 - e^{-\pi s})(s^2 + 1)}.$$

Multiply numerator and denominator by $e^{\pi s/2}$:

$$\frac{1 + e^{-\pi s}}{1 - e^{-\pi s}} = \frac{e^{\pi s/2} + e^{-\pi s/2}}{e^{\pi s/2} - e^{-\pi s/2}} = \coth\left(\frac{\pi s}{2}\right).$$

$$\mathcal{L}\{|\sin t|\} = \frac{\coth(\pi s/2)}{s^2 + 1}.$$

What Did We Learn?

The full-wave rectified sine has half the period of the half-wave version, giving a cleaner coth expression. The formula $\coth(\pi s/2)/(s^2 + 1)$ appears frequently in power electronics Laplace analysis.

10 Excel Example

Objective. Numerically verify the square wave formula $\mathcal{L}\{f\} = \tanh(0.5)/1$ at $s = 1$, $a = 1$, $A = 1$ (exact value ≈ 0.4621) using the trapezoidal rule.

Column A	Column B	Column C	Column D
t	f(t)	$e^{-(t)} \cdot f(t)$	Cumulative Trapezoid
0.0	=IF(MOD(A2,2)<1,1,-1)	=EXP(-1*A2)*B2	0
0.1	=IF(MOD(A3,2)<1,1,-1)	=EXP(-1*A3)*B3	=D2+(C2+C3)*0.1/2
0.2	=IF(MOD(A4,2)<1,1,-1)	=EXP(-1*A4)*B4	=D3+(C3+C4)*0.1/2
...	...	...	(drag down to row 101: t=10)
10.0	=IF(MOD(A102,2)<1,1,-1)	=EXP(-1*A102)*B102	=D101+(C101+C102)*0.1/2

Setup instructions:

1. In cell A2 enter 0; in A3 enter =A2+0.1; drag A3 down to A102 ($t = 0.0$ to 10.0).
2. Column B: =IF(MOD(A2,2)<1,1,-1) detects which half-period we are in (+1 if fractional part of t in $[0, 2)$ is less than 1, else -1).
3. Column C: =EXP(-1*A2)*B2 computes $e^{-t}f(t)$ at each point.
4. Column D row 2: enter 0; from row 3 onward: =D2+(C2+C3)*0.1/2 (trapezoidal rule).

5. Cell D102 gives the numerical approximation to $\int_0^{10} e^{-t} f(t) dt$.
6. The integral $\int_0^{10} e^{-t} f(t) dt$ directly approximates the Laplace transform $\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-t} f(t) dt = \frac{1}{s} \tanh(as/2) = \tanh(0.5) \approx 0.4621$ (for $s = 1, a = 1$). The periodic theorem gives $\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2}} \int_0^2 e^{-t} f(t) dt$, where the single-period integral is $\int_0^2 e^{-t} f(t) dt = (1 - e^{-2}) \tanh(0.5)$. Both yield the exact value $\tanh(0.5) \approx 0.4621$.
7. In cell F2 enter the exact formula: `=TANH(0.5)/1` to display 0.4621.
8. Compare D102 with F2; the difference should be less than 0.002 (small because $e^{-10} < 0.00005$).

What Did We Learn?

The numerical integration in Excel converges to $\tanh(0.5) \approx 0.4621$, confirming the analytic formula. Truncating the infinite integral at $t = 10$ introduces an error of order $e^{-10} \approx 4.5 \times 10^{-5}$, which is negligible.

11 Python Example

Listing 1: Numerical verification of the square wave Laplace transform

```

1 import numpy as np
2 from scipy.integrate import quad
3
4 # Parameters
5 A = 1.0    # amplitude
6 a = 1.0    # half-period; full period T = 2a = 2
7 T = 2 * a
8 s = 1.0    # Laplace variable
9
10 # Define the square wave (one cycle repeated)
11 def square_wave(t):
12     # positive half [0, a), negative half [a, 2a)
13     phase = t % T
14     return A if phase < a else -A
15
16 # Numerically integrate e^{-st} f(t) from 0 to infinity (approx 0
    to 50)
17 integrand = lambda t: np.exp(-s * t) * square_wave(t)
18 direct_integral, err = quad(integrand, 0, 50, limit=500)
19
20 # One-period integral
21 one_period, _ = quad(integrand, 0, T, limit=200)
22
23 # Periodic theorem formula: F(s) = one_period / (1 - exp(-sT))
24 formula_value = one_period / (1.0 - np.exp(-s * T))
25
26 # Exact closed-form: (A/s) * tanh(a*s/2)
27 exact_value = (A / s) * np.tanh(a * s / 2)

```

```

28
29 print("=== Square Wave Laplace Transform at s=1 ===")
30 print(f"Direct numerical integral (0 to 50): {direct_integral:.6f
31       }")
32 print(f"Periodic theorem formula           : {formula_value:.6f}
33       ")
34 print(f"Closed-form (A/s)*tanh(as/2)       : {exact_value:.6f}")
35 print(f"Error (direct vs exact)             : {abs(
36       direct_integral - exact_value):.2e}")
37 print(f"Error (formula vs exact)           : {abs(formula_value
38       - exact_value):.2e}")
39
40 # Expected output:
41 # === Square Wave Laplace Transform at s=1 ===
42 # Direct numerical integral (0 to 50): 0.462117
43 # Periodic theorem formula           : 0.462117
44 # Closed-form (A/s)*tanh(as/2)       : 0.462117
45 # Error (direct vs exact)             : 4.55e-05
46 # Error (formula vs exact)           : 1.11e-16

```

What Did We Learn?

The direct numerical integral, the periodic theorem formula, and the closed form $\tanh(0.5)/1 \approx 0.4621$ all agree to six decimal places. The periodic theorem formula achieves machine-epsilon accuracy, while the direct integral has a small but bounded error from truncating at $t = 50$.

12 Viva-Style Oral Questions

Q1. Define a periodic function.

Answer. A function $f(t)$ is periodic with period $T > 0$ if $f(t + T) = f(t)$ for all $t \geq 0$. The smallest such T is the fundamental period.

Q2. State the Laplace transform formula for a periodic function.

Answer.

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt, \quad s > 0.$$

Q3. How does the geometric series arise in the proof of the periodic function theorem?

Answer. After substituting $t = u + nT$ in each n -th piece of the definition integral, periodicity gives $f(u + nT) = f(u)$, and the factor e^{-snT} comes outside. Summing over $n \geq 0$ produces $\sum_{n=0}^{\infty} (e^{-sT})^n = 1/(1 - e^{-sT})$, which is a geometric series converging for $|e^{-sT}| < 1$, i.e., $s > 0$.

Q4. What does the factor $1/(1 - e^{-sT})$ represent physically?

Answer. It is the sum of an infinite geometric series with ratio e^{-sT} , accounting for every repeated cycle of the waveform. In financial terms it is the “present value multiplier” for infinitely many periodic payments discounted at rate s .

Q5. How do you correctly identify the period T of a piecewise periodic function?

Answer. T is the smallest positive number such that the piecewise definition repeats exactly. For a square wave defined on $[0, a)$ and $[a, 2a)$, the period is $T = 2a$, *not* a . Choosing $T = a$ (half-period) is the most common error and gives a completely wrong integral.

Q6. What is the Laplace transform of a symmetric square wave with amplitude A and half-period a ?

Answer.

$$\mathcal{L}\{f(t)\} = \frac{A}{s} \tanh\left(\frac{as}{2}\right).$$

Q7. Why is direct integration impractical for periodic functions?

Answer. The definition integral $\int_0^\infty e^{-st} f(t) dt$ must be split into infinitely many separate integrals (one per period), each evaluated individually, and then summed. This infinite computation cannot be performed analytically in closed form without recognising the geometric series structure, and is totally impractical in an exam context.

Q8. Describe the application of the periodic function theorem to the half-wave rectifier.

Answer. The half-wave rectified sine has period $T = 2\pi$ and equals $\sin t$ on $[0, \pi)$ and 0 on $[\pi, 2\pi)$. Only the $[0, \pi]$ piece contributes to the one-period integral. After integrating $\int_0^\pi e^{-st} \sin t dt = (1 + e^{-\pi s})/(s^2 + 1)$ and dividing by $(1 - e^{-2\pi s})$, we cancel $(1 + e^{-\pi s})$ to obtain $1/[(1 - e^{-\pi s})(s^2 + 1)]$.

13 Descriptive Questions

1. State and prove the Laplace transform theorem for periodic functions.

State the theorem precisely, split the definition integral over successive periods of length T , apply the substitution $t = u + nT$, use periodicity to replace $f(u + nT)$ with $f(u)$, factor out e^{-snT} , and sum the resulting geometric series $\sum_{n=0}^\infty e^{-snT} = 1/(1 - e^{-sT})$ to obtain the formula.

2. Find the Laplace transform of the square wave with amplitude A and half-period a .

Define the square wave on $[0, 2a)$, apply the theorem with $T = 2a$, evaluate the two-piece integral, combine and factor, and simplify using the tanh identity to obtain $(A/s) \tanh(as/2)$.

3. Find the Laplace transform of the triangular wave with half-period a .

Define the triangular wave on $[0, 2a)$, apply the theorem, evaluate the two integration-by-parts integrals for the linear pieces, combine to obtain $(1 - e^{-as})^2/s^2$, divide by $(1 - e^{-2as})$, and simplify to $(1/s^2) \tanh(as/2)$.

4. Derive the Laplace transform of the half-wave rectified sine $f(t) = \sin t$ on $[0, \pi)$, 0 on $[\pi, 2\pi)$.

Apply the theorem with $T = 2\pi$; only $\int_0^\pi e^{-st} \sin t dt$ contributes. Evaluate by two

rounds of integration by parts to get $(1 + e^{-\pi s})/(s^2 + 1)$, divide by $(1 - e^{-2\pi s})$, and cancel $(1 + e^{-\pi s})$ to obtain $1/[(1 - e^{-\pi s})(s^2 + 1)]$.

5. Explain the connection between the geometric series and the Laplace transform of a periodic function.

Show how splitting the integral at $t = nT$ and applying the substitution $t = u + nT$ produces a common factor $\int_0^T e^{-su} f(u) du$ multiplied by e^{-snT} for each n . Summing $\sum_{n=0}^{\infty} e^{-snT}$ as a geometric series with ratio $r = e^{-sT} < 1$ yields $1/(1 - e^{-sT})$, so the full transform is this factor times the one-period integral.

14 Practice Problems

P1. Find $\mathcal{L}\{f(t)\}$ where $f(t) = E$ for $0 \leq t < T/2$, $f(t) = 0$ for $T/2 \leq t < T$, period T (rectangular pulse wave).

Hint: $\frac{E}{s} \cdot \frac{1 - e^{-sT/2}}{1 - e^{-sT}} = \frac{E}{s(1 + e^{-sT/2})}$.

P2. A staircase function is defined by $f(t) = n$ for $nT \leq t < (n+1)T$, $n = 0, 1, 2, \dots$. This is *not* periodic in the usual sense, but can be written as a sum of step functions. Express $\mathcal{L}\{f(t)\}$ in terms of e^{-sT} .

Hint: Write $f(t) = \sum_{n=1}^{\infty} u(t - nT)$; then $\mathcal{L}\{f(t)\} = \frac{e^{-sT}}{s(1 - e^{-sT})}$.

P3. Find $\mathcal{L}\{|\cos t|\}$ (full-wave rectified cosine, period π).

Hint: $\frac{s \coth(\pi s/2)}{s^2 + 1}$.

P4. A sawtooth wave has $f(t) = 2t/T$ for $0 \leq t < T$ (slope $2/T$), period T . Find $\mathcal{L}\{f(t)\}$.

Hint: Similar to Example 1; result is $\frac{2}{Ts^2} - \frac{2}{s(e^{sT} - 1)}$.

P5. A periodic function is described as $f(t) = \sin(\omega t)$ with “period π/ω ”. Identify the error and state the correct period.

Hint: The fundamental period of $\sin(\omega t)$ is $T = 2\pi/\omega$, not π/ω . Using the wrong period would apply the formula over only a half-cycle.

P6. Use the tanh identity to verify that $(A/s)(1 - e^{-as})/(1 + e^{-as})$ equals $(A/s) \tanh(as/2)$ by multiplying numerator and denominator by $e^{as/2}$.

Hint: $\frac{e^{as/2} - e^{-as/2}}{e^{as/2} + e^{-as/2}} = \tanh(as/2)$ by definition of tanh.

15 MCQs

MCQ 1. The Laplace transform of a periodic function $f(t)$ with period T is:

(a) $\int_0^T e^{-st} f(t) dt$

- (b) $\frac{1}{1 - e^{sT}} \int_0^T e^{-st} f(t) dt$
- (c) $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$
- (d) $(1 - e^{-sT}) \int_0^T e^{-st} f(t) dt$

Explanation: Option (c) is the correct periodic function theorem. The factor $1/(1 - e^{-sT})$ accounts for all periods; dropping it (option a) gives only the first-period contribution.

MCQ 2. The denominator factor $1 - e^{-sT}$ in the periodic Laplace formula arises from:

- (a) Integration by parts applied to $e^{-st} f(t)$
- (b) **Summing the geometric series** $\sum_{n=0}^{\infty} e^{-snT}$
- (c) Partial fraction decomposition of $F(s)$
- (d) The initial value theorem

Explanation: After substituting $t = u + nT$ in each piece of the definition integral, one sums $\sum e^{-snT} = (1 - e^{-sT})^{-1}$, a geometric series.

MCQ 3. A function is defined as $f(t) = 1$ for $0 \leq t < 2$ and $f(t) = -1$ for $2 \leq t < 4$, with $f(t + 4) = f(t)$. The correct period T to use in the formula is:

- (a) $T = 1$
- (b) $T = 2$
- (c) $T = 4$
- (d) $T = 8$

Explanation: The function completes one full cycle (positive then negative) over $[0, 4)$, so the fundamental period is $T = 4$. Using $T = 2$ would apply the formula over only the positive half.

MCQ 4. The Laplace transform of a square wave with amplitude $A = 3$ and half-period $a = 2$ is:

- (a) $\frac{3}{s} \coth(s)$
- (b) $\frac{3}{s} \sinh(s)$
- (c) $\frac{3}{s} \tanh(s)$
- (d) $\frac{3}{s^2} \tanh(s)$

Explanation: Using $A = 3$, $a = 2$: $(A/s) \tanh(as/2) = (3/s) \tanh(s)$. Option (d) is wrong because it has s^2 in the denominator (that belongs to the triangular wave).

MCQ 5. As $T \rightarrow \infty$ in the periodic function formula, the factor $1/(1 - e^{-sT})$ approaches:

- (a) 0
- (b) ∞
- (c) 1
- (d) e^{sT}

Explanation: As $T \rightarrow \infty$, $e^{-sT} \rightarrow 0$ for $s > 0$, so $1/(1 - e^{-sT}) \rightarrow 1/(1 - 0) = 1$. The formula reduces to the standard definition integral, as expected for a non-periodic (“period-infinite”) function.

16 Common Mistakes

Common Mistakes in Periodic Function Problems

Mistake	What Students Do	Correct Approach
Forgetting the $1/(1 - e^{-sT})$ factor	Treat $\mathcal{L}\{f\}$ as just the one-period integral	Always multiply the one-period integral by $1/(1 - e^{-sT})$
Using the wrong period T	Use a (half-period) instead of $2a$ for a square wave	Identify T as the length of one complete cycle, not one half
Not checking piecewise continuity	Apply the theorem to a function with jump discontinuities exceeding what the theorem allows	Verify $f(t)$ is piecewise continuous on $[0, T]$ before applying the theorem
Incorrect limits in the one-period integral	Integrate from 0 to ∞ instead of 0 to T	The upper limit in the formula is always T (one period only)
Forgetting the sign of the negative half	Write $f(t) = A$ on both $[0, a)$ and $[a, 2a)$	The square wave is $+A$ on $[0, a)$ and $-A$ on $[a, 2a)$; incorrect sign changes the entire result

17 Quick Recap

Quick Recap — Topic 06: Laplace Transform of Periodic Functions

- **Periodic function:** $f(t + T) = f(t)$ for all $t \geq 0$; fundamental period is the smallest such T .
- **Periodic function theorem:** $\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt, s > 0$.
- **Proof strategy:** Split the definition integral at $t = nT$, substitute $t = u + nT$, apply periodicity $f(u + nT) = f(u)$, and sum the geometric series $\sum_{n=0}^{\infty} e^{-snT} = 1/(1 - e^{-sT})$.

- **Square wave result:** Amplitude A , half-period a : $\mathcal{L}\{f(t)\} = \frac{A}{s} \tanh\left(\frac{as}{2}\right)$.
- **Triangular wave result:** Half-period a : $\mathcal{L}\{f(t)\} = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right)$.
- **Half-wave rectifier:** $f(t) = \sin t$ on $[0, \pi)$, zero on $[\pi, 2\pi)$: $\mathcal{L}\{f(t)\} = \frac{1}{(1 - e^{-\pi s})(s^2 + 1)}$.
- **Financial analogy:** The factor $1/(1 - e^{-sT})$ is the present-value multiplier for infinitely many recurring “payments” (one-period integrals), discounted at rate s per period.
- **Key caution:** Always use the full period T (not half-period) as the upper limit of integration; the $\tanh$ simplification for symmetric waves follows from $(1 - e^{-as})/(1 + e^{-as}) = \tanh(as/2)$.